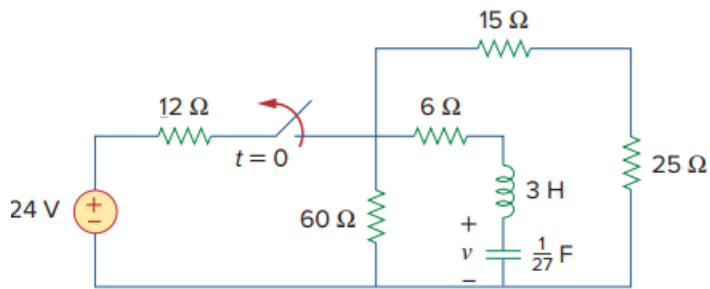


Problem

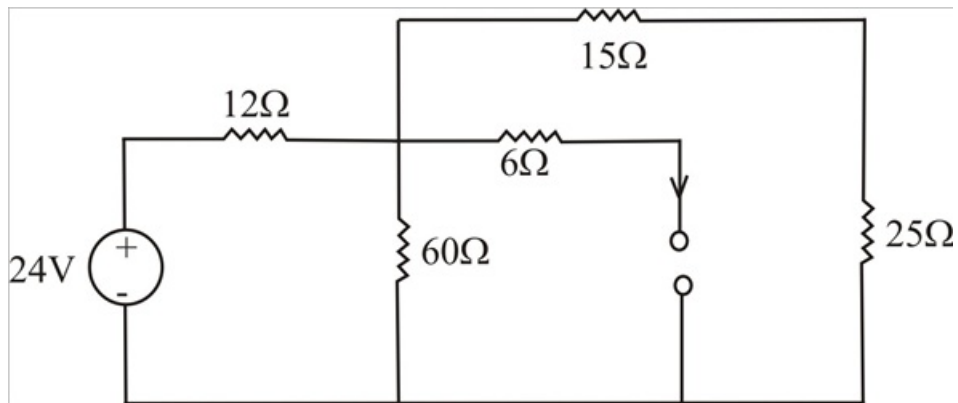
Calculate $v(t)$ for $t > 0$ in the circuit of Fig. 8.75.

Figure 8.75:



Step-by-step solution

Step 1 of 6



Step 2 of 6

At $t = 0^-$,

$$V_C(0^+) = V_C(0^-) \quad \text{And} \quad i_L(0^+) = i_L(0^-) = 0$$

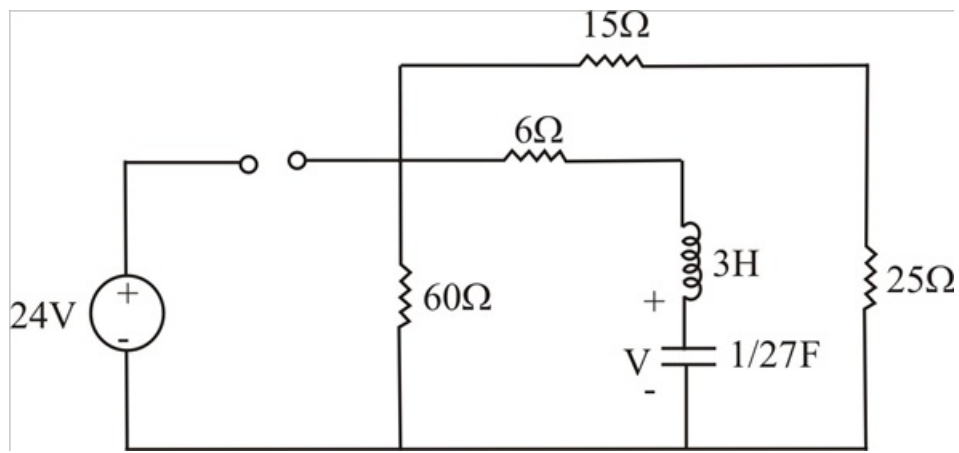
$$\text{Total current} = \frac{24}{12 + [(15 + 25) \parallel 60]} = \frac{24}{12 + \left(\frac{2400}{100}\right)} = \frac{2}{3} A$$

$$\text{So, } I_{60\Omega} = \frac{2}{3} \times \frac{40}{100} = \frac{4}{15} A$$

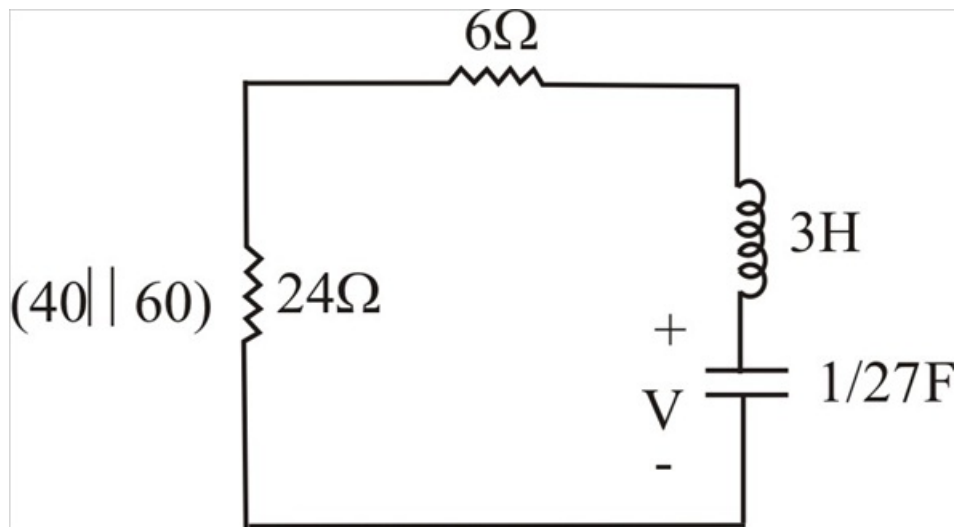
$$\text{So, } V_{60\Omega} = \frac{4}{15} \times 60\Omega = 16V$$

$$V_C(0^-) = 16V$$

Step 3 of 6



Step 4 of 6



Step 5 of 6

At $t = 0+$,

$$\alpha = \frac{R}{2L} = \frac{30}{6} = 5$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 3 \quad [C = \frac{1}{27}F ; L = 3H]$$

$\therefore \alpha > \omega_0$ [Overdamped]

$$\text{So, } V(t) = Ae^{S_1 t} + Be^{S_2 t} \rightarrow (1)$$

$$S_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -5 + \sqrt{25 - 9} = -1$$

$$S_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -5 - \sqrt{25 - 9} = -9$$

$$\text{Or, } V(t) = Ae^{-t} + Be^{-9t}$$

Step 6 of 6

At $t = 0$;

$$V(0) = 16V = A + B$$

And,

$$\frac{dV(t)}{dt} = -Ae^{-t} - 9Be^{-9t}$$

$$\Rightarrow \frac{dV(0)}{dt} = 0 = -A - 9B$$

On solving we get $A = 18$; $B = -2$;

$$\text{So, } V(t) = 18e^{-t} - 2e^{-9t} V$$